

Recent Progress on Using AI for Research in Statistics

Edgar Dobriban

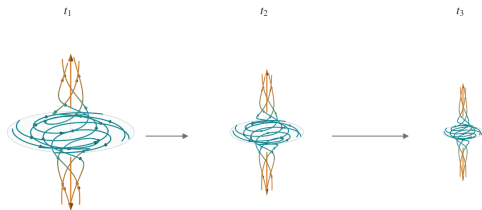
Department of Statistics and Data Science, the Wharton School, University of Pennsylvania

September 23, 2026



FINITE TIME BLOWUP FOR NAVIER-STOKES

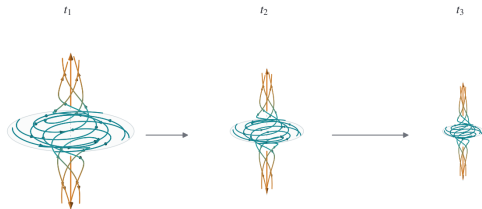
OPENAI



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FINITE TIME BLOWUP FOR NAVIER-STOKES

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They used 10,000 agents powered by an internal model (costing cca \$20M), and released a proof and a Lean formalization (600,000 lines of code).

An N.Y.U. Mathematician Clashed With OpenAI Over a \$1 Million Proof

NYU mathematician Tristan Buckmaster (working on same problem w/ collaborator T. Alpöge over last year using in part OpenAI models) raised concerns about scientific credit.

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OpenAI responded, controversy ongoing.

Beyond excitement, a mixed reaction from the community

A Severe Misalignment of AI in Mathematics

11 September, 2026 in [advertising](#), [opinion](#) | Tags: [AI](#), [Fields medals](#) | by [Terence Tao](#)

Math and AI.



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On September 11, Terence Tao introduced a declaration with 25 initial signatories, all Fields medalists.

- ▶ The letter argues that the incentives of AI companies for solving problems are misaligned with those of the mathematical community. AI companies use math problems as advertising.
- ▶ The math community cares deeply about understanding, proper attribution, and community.

Feedback from the experts

The slide below is from a talk by Javier Gomez-Serrano (top expert on Navier-Stokes, one of the researchers whose recent work the solution crucially builds on), see

<https://www.youtube.com/watch?v=TcEefrWrddA>.

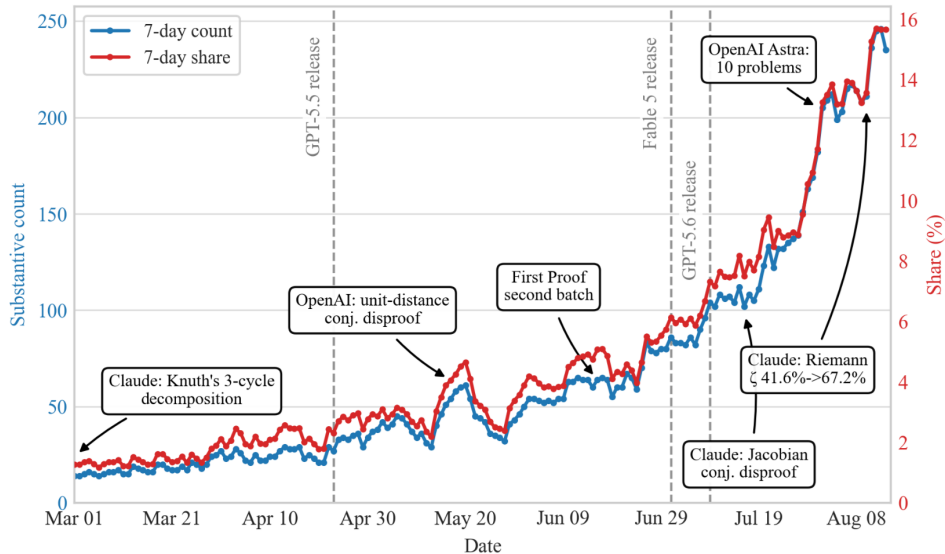
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A message to OpenAI (and to everyone building these systems)

- ▶ Rewrite the paper in a more comprehensible way, perhaps with the help and/or advice of mathematicians trained in mathematical fluid mechanics.
- ▶ Release the full process: prompts, transcripts, intermediate attempts, the Lean project with its statement audit.
- ▶ Cite generously and early.
- ▶ Give the community time, compute, and access, not only announcements.
- ▶ If you want to accelerate mathematical research, more researchers need to have access to state of the art tools.

Bigger picture: a "gold rush" (paper by Jin, Ke, Sui)



Vector Balancing via Directional Total Variation

Shengtao Guo

Ethan X. Fang

Junwei Lu*

Abstract

Our main result is a $3\sqrt{2\pi}$ bound for the Komlós signing problem: every finite family of real vectors of Euclidean norm at most one admits a signed sum of ℓ_∞ -norm less than this constant, independently of the dimension and the family size.

Theorem 1.1. *For every pair of positive integers m, n and every $v_1, \dots, v_n \in \mathbb{R}^m$ with $\|v_j\|_2 \leq 1$, there are $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ such that*

$$\left\| \sum_{j=1}^n \varepsilon_j v_j \right\|_\infty < 3\sqrt{2\pi}. \quad (1.1)$$

The Berry–Esseen Bound is Sharp for All Sufficiently Large Sample Sizes

Hengzhi He* Guang Cheng*

September 2026

Abstract

We prove that the Esseen’s constant 0.4097... is sharp in the iid Berry–Esseen inequality as long as the sample sizes $n \geq N$. The threshold N is universal over all real summand distributions with positive variance and a finite third absolute moment. In particular, one may take $N = 2 \lceil \exp(10^{17}) \rceil$.

1 Introduction

Let $X_1, X_2, \dots$ be independent identically distributed real random variables with mean zero, variance one and finite third absolute moment $\beta = \mathbb{E}|X_1|^3$. The Berry–Esseen theorem asserts that

$$\sup_{x \in \mathbb{R}} \left| \mathbb{P} \left\{ \frac{X_1 + \dots + X_n}{\sqrt{n}} \leq x \right\} - \Phi(x) \right| \leq C \frac{\beta}{\sqrt{n}} \quad (1)$$

But what about statistical research itself? Early work

arXiv:2511.18828v1 [math.ST] 24 Nov 2025

Solving a Research Problem in Mathematical Statistics with AI Assistance

Edgar Dobriban*

November 25, 2025

Abstract

Over the last few months, AI models including large language models have improved greatly. There are now several documented examples where they have helped professional mathematical scientists prove new results, sometimes even helping resolve known open problems. In this short note, we add another example to the list, by documenting how we were able to solve a previously unsolved research problem in robust mathematical statistics with crucial help from GPT-5. Our problem concerns robust density estimation, where the observations are perturbed by Wasserstein-bounded contaminations. In a previous preprint (Chao and Dobriban, 2023, arxiv:2308.01853v2), we have obtained upper and lower bounds on the minimax optimal estimation error; which were, however, not sharp.

Starting in October 2025, making significant use of GPT-5 Pro, we were able to derive the minimax optimal error rate (reported in version 3 of the above arxiv preprint). GPT-5 provided crucial help along the way, including by suggesting calculations that we did not think of, and techniques that were not familiar to us, such as the dynamic Benamou-Brenier formulation, for key steps in the analysis. Working with GPT-5 took a few weeks of effort, and we estimate that it could have taken several months to get the same results otherwise. At the same time, there are still areas where working with GPT-5 was challenging: it sometimes provided incorrect references, and glossed over details that sometimes took days of work to fill in. We outline our workflow and steps taken to mitigate issues. Overall, our work can serve as additional documentation for a new age of human-AI collaborative work in mathematical science.

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- ▶ Human-only (Feb. 2021 - Aug. 2023):

$$\max \left\{ n^{-2s/(2s+1)}, \varepsilon^{2s/(s+1/2)} \right\} \lesssim \mathcal{M}(\varepsilon, s, 1, 2, 2) \lesssim \max \left\{ n^{-2s/(2s+1)}, \varepsilon^{2s/(s+2)} \right\}.$$

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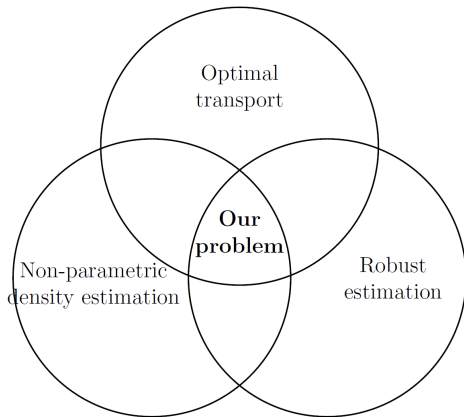
$$\max \left\{ n^{-2s/(2s+1)}, \varepsilon^{\frac{2s}{s+1/2}} \right\} \lesssim \mathcal{M}(\varepsilon, s, 1, 2, 2) \lesssim \max \left\{ n^{-2s/(2s+1)}, \varepsilon^{\frac{2s}{s+2}} \right\}.$$

- ▶ AI-assisted (Oct. - Nov. 2025):

$$\mathcal{M}(\varepsilon, s, p, q, r) \asymp \max \left\{ n^{-\frac{2s}{2s+p}}, \varepsilon^{\frac{2s}{s+1+p/r}} \right\}.$$

$$\text{In particular, } \mathcal{M}(\varepsilon, s, 1, 2, 2) \asymp \max \left\{ n^{-\frac{2s}{2s+1}}, \varepsilon^{\frac{2s}{s+3/2}} \right\}.$$

Solving a Statistics Research Problem with AI Help (ctd)



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In our **lower bound**, we have adapted the modulus of continuity approach, by starting with a normal density $f = \varphi_\sigma$ with sufficiently small variance $\sigma^2 > 0$, and perturbing it with a mean zero function to obtain a contaminated density $\tilde{f}$. Then, we upper bounded $W_{2,2}(f, \tilde{f}) \leq \varepsilon$ (via the Talagrand transportation cost inequality, see Talagrand (1996); Otto and Villani (2000)) and lower bounded $|f(x_0) - \tilde{f}(x_0)|$, which—by the modulus of continuity approach—lead to a lower bound on the estimation error.

Solving a Statistics Research Problem with AI Help (ctd)

GPT-5 Suggests Dynamic Benamou-Brenier for Improved Lower Bound (Extract)

User: [Prompt includes LaTeX draft of paper and a query: "Can you go through this proof and see if you can improve either the upper bound and the lower bound until they match?"]

GPT: [...]

Lemma [..] scales like $W_2^2 \lesssim D^2 \int g_+$, i.e. it is linear in the bump amplitude τ while the correct small-amplitude scaling for W_2 in one dimension is quadratic in τ .

[...] We state and prove the replacement for Lemma [..].

Lemma (Refined local W_2 bound via the dynamic formulation)

Let q be a probability density on $\mathbb{R}$ and let $I \subset \mathbb{R}$ be a finite interval. Let $g \in L^1(\mathbb{R})$ satisfy $\text{supp}(g) \subset I$ and $\int g = 0$. Write $G(x) = \int_{-\infty}^x g(t) dt$. Suppose there exists $c_I > 0$ such that $q(x) \geq c_I$ on I and $|g|_\infty \leq c_I/2$. Then $q + g$ is a density and

$$W_2^2(q + g, q) \leq 2c_I \int_{\mathbb{R}} G(x)^2 dx.$$

Solving a Statistics Research Problem with AI Help (ctd)

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Proof.

Consider the path of densities $\rho_t = q + tg$ for $t \in [0, 1]$. On I we have $\rho_t \geq q - |g|_\infty \geq c_I/2$, while off I one has $\rho_t = q > 0$. Define $v_t(x) = -G(x)/\rho_t(x)$. Then $\rho_t v_t = -G$ is independent of t , hence $\partial_x(\rho_t v_t) = -\partial_x G = -g$ and $\partial_t \rho_t = g$; thus $\partial_t \rho_t + \partial_x(\rho_t v_t) = 0$. Applying the Benamou-Brenier dynamic formulation gives

$$W_2^2(q + g, q) \leq \int_0^1 \int_{\mathbb{R}} |v_t(x)|^2 \rho_t(x) dx dt = \int_0^1 \int_{\mathbb{R}} G(x)^2 dx dt = \int_{\mathbb{R}} G(x)^2 dx.$$

Since $\rho_t(x) \geq c_I/2$ on I and $\rho_t = q$ off I , the bound sharpens to

$$W_2^2(q + g, q) \leq 2c_I \int_{\mathbb{R}} G(x)^2 dx.$$



But what about statistical research itself? Recent work

An example

AI helps understand the limitations of a widely used method for multiple hypothesis testing

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AI helps understand the limitations of a widely used method for multiple hypothesis testing

A case study

Develop a new estimator for variance estimator in non-parametric regression.

The Benjamini–Hochberg Procedure Can Fail to Control the FDR for Correlated Two-Sided Gaussian Tests

Edgar Dobriban*

July 15, 2026

Abstract

We show that the Benjamini–Hochberg procedure can fail to control the false discovery rate (FDR) at its nominal level for correlated two-sided Gaussian p -values. We construct a factor model for which, at level $\alpha = 0.01$, a rigorous interval-arithmetic certificate proves $\text{FDR} > 0.0104$ for all sufficiently large numbers of hypotheses. This disproves a conjecture widely believed to be true for twenty years. Monte Carlo experiments are consistent with the theoretical result. The proof was obtained by GPT-5.6 Pro and carefully checked by the author.

False discovery rate

Multiple testing notation

Suppose we test m null hypotheses. Let

$R = \{\text{rejected hypotheses}\}$, $V = \{\text{false rejections}\}$.

Here V is the number of discoveries that are actually null, while R is the total number of discoveries.

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Controlling the false discovery rate: a practical and powerful approach to multiple testing

[Y Benjamini](#), Y Hochberg - Journal of the Royal statistical ..., 1995 - Wiley Online Library

... From this point of view, a desirable error **rate** to **control** may be the expected proportion of errors among the rejected hypotheses, which we term the **false discovery rate** (FDR). This ...

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Does the Benjamini–Hochberg procedure control the false discovery rate at level α , i.e., do we have $\text{FDR} \leq \alpha$, for every m , every mean vector, every correlation matrix, and every $\alpha \in (0, 1)$?

Before this work, a positive answer was widely believed. [Farcomeni \(2006\)](#); [Reiner-Benaim \(2007\)](#) provided empirical and theoretical evidence in support of FDR control, and [Kim and van de Wiel \(2008\)](#) provided additional evidence through extensive simulations. In an influential review, [Benjamini \(2010\)](#) wrote that “*convincing simutheoretical evidence indicates that [FDR control] holds for two-sided z-tests with any correlation structure.*” This motivates referring to the positive answer as a conjecture.

More recently, [Sarkar \(2023\)](#) wrote that “*The answer to this question is generally believed to be yes, and is conjectured so in the literature since results of numerical studies investigating the question and reported in numerous papers strongly support it.*” [Sarkar \(2023\)](#) continued that “*proving this conjecture [...] seems an urgent and important undertaking.*” Similar statements were made by [Sarkar and Zhang \(2025\)](#) and [Ghosh and Sarkar \(2025\)](#).

In this paper we show that the conjecture fails, by constructing the following Gaussian factor model. Let Z , $(\varepsilon_i)_{i \geq 1}$, $(\eta_j)_{j \geq 1}$, $(\xi_k)_{k \geq 1}$ be mutually independent standard normal random variables. For a fixed N , define three coordinate blocks by

$$X_i^{(0)} = \frac{3}{10}Z + \frac{\sqrt{91}}{10}\varepsilon_i, \quad 1 \leq i \leq 96N, \quad (1)$$

$$X_j^{(1)} = \frac{12}{5} - \frac{3}{10}Z + \frac{\sqrt{91}}{10}\eta_j, \quad 1 \leq j \leq N, \quad (2)$$

$$X_k^{(2)} = \frac{22}{5} - \frac{18}{25}Z + \frac{\sqrt{301}}{25}\xi_k, \quad 1 \leq k \leq 3N. \quad (3)$$

The block in (1) contains the $96N$ true nulls. The other two blocks contain the $4N$ nonnulls. The null block moves with Z , whereas both signal blocks move against it, at two different strengths.

Theorem 1 (The Benjamini–Hochberg Procedure Can Fail to Control the FDR for Correlated Two-Sided Gaussian Tests). *For each integer $N \geq 1$, let $m_N = 100N$, let the first $96N$ hypotheses be true nulls, and let FDR_N denote the FDR in the model above. Then the Benjamini–Hochberg procedure at level $\alpha = 0.01$ satisfies, for all sufficiently large N ,*

$$\text{FDR}_N > 0.0104 > \alpha.$$

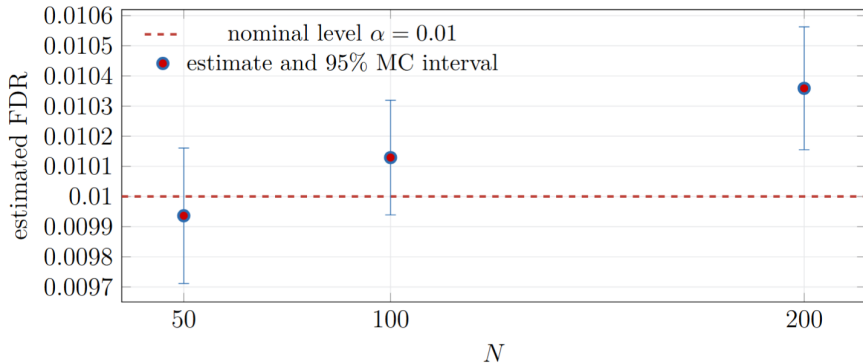


Figure 4: Finite-sample stratified Monte Carlo estimates. The first two intervals do not resolve the sign of $\text{FDR}_N - \alpha$, while the interval at $N = 200$ lies wholly above the nominal level.

Prove or disprove the following conjecture about BH for two-sided Gaussian tests:

For $X \sim N_m(\mu, \Sigma)$ with $\Sigma_{ii} = 1$, testing $H_0: \mu = 0$ against two-sided alternatives with $P_i = 2\bar{\Phi}(|X_i|)$, the ordinary BH procedure at level α satisfies $\text{FDR} \leq \alpha$ for every μ , every correlation matrix Σ , and every $\alpha \in [0, 1]$.

Use the following conventions.

Show more 

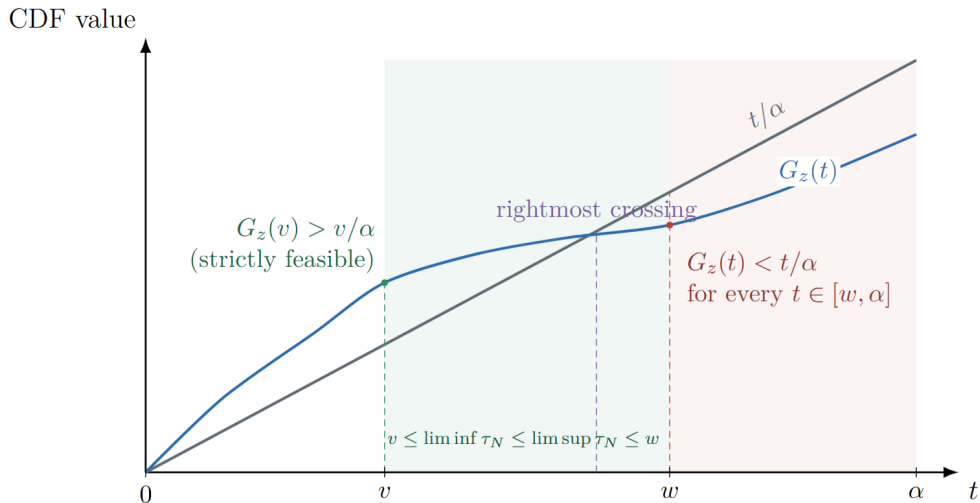
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The conjecture is false

There is an explicit three-block Gaussian factor model for which ordinary BH at level $\alpha = 0.01$ has

$$\liminf_{N \rightarrow \infty} \text{FDR}_N > 0.0104168290704 > 0.01.$$

The proof builds on ideas from the standard asymptotic analysis of FDR (Genovese and Wasserman, 2002, 2004; Storey et al., 2004).





Marc Andreessen 🇺🇸 🗳️ 📺 @pmarca · Jul 17



Amazing!



Edgar Dobriban @EdgarDobriban · Jul 14

AI has helped resolve an important question in statistics. In the area of multiple hypothesis testing, the goal of controlling the false discovery rate (FDR) has been introduced in a seminal paper by Benjamini and Hochberg (1995). They also introduced a method (the

Benjamini-Hochberg Procedure Can Fail to Control the FDR for Correlated Two-Sided Gaussian Factor Model

Edgar Dobriban*

July 14, 2026

Abstract

At the Benjamini-Hochberg procedure can fail to control the FDR at its nominal level for correlated two-sided Gaussian factor model for which, at level $\alpha = 0.01$, a rigorous inference shows $\text{FDR} > 0.0104$ for all sufficiently large numbers of hypotheses. This is a conjecture widely believed to be true for twenty years, but is now inconsistent with the theoretical result. The proof has been carefully checked by the author.

factor model. Let $Z, (\zeta_i)_{i=1, \dots, 90}, (\eta_j)_{j=1, \dots, 10}, (\zeta_k)_{k=1, \dots, 3}$ be mutually independent standard normal random variables. For a fixed N , define three coordinate blocks by

$$X_i^{(0)} = \frac{3}{10}Z + \frac{\sqrt{91}}{10}\zeta_i, \quad 1 \leq i \leq 90N, \quad (1)$$

$$X_j^{(1)} = \frac{12}{5} - \frac{3}{10}Z + \frac{\sqrt{91}}{10}\eta_j, \quad 1 \leq j \leq N, \quad (2)$$

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The block in (1) contains the $90N$ true nulls. The other two blocks contain the $4N$ nonnulls. The null block moves with Z , whereas both signal blocks move against it, at two different strengths.

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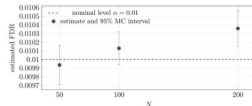


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31

31

254

109K



How Much Can Gaussian Dependence Inflate the Benjamini–Hochberg Procedure’s FDR?

Lihua Lei

Graduate School of Business, Stanford University

lihual lei@stanford.edu

August 5, 2026

Abstract

We study the worst-case false discovery rate (FDR) of the Benjamini–Hochberg procedure for both one- and two-sided Gaussian tests when the correlation matrix is otherwise unrestricted. In each setting we construct a q -indexed family of finite Gaussian models whose FDR divided by q diverges as $q \downarrow 0$, disproving any universal multiplicative FDR bound. For two-sided tests, the supremum over the number of hypotheses, mean vector, and correlation matrix is at least an explicit $\ell_=(q) > q$ satisfying

$$\ell_=(q) = \frac{q\sqrt{\log(1/q)}}{2\sqrt{\pi}} + c_\ell q + o(q), \quad c_\ell = 0.6492828 \dots$$

For the one-sided hypotheses $H_i : \theta_i \leq 0$, a sign-reversed one-common-factor construction gives the stronger explicit lower bound $\ell_\leq(q) > q$, with

$$\ell_\leq(q) = \frac{q\sqrt{\log(1/q)}}{\sqrt{\pi}} + \frac{q}{2} + o(q).$$

Second example: variance estimation

Improved Variance Estimation in Homoskedastic Nonparametric Random-Design Regression via a Two-Scale Approach

Edgar Dobriban, Rajarshi Mukherjee, James M. Robins, and Zixiao Wang*

September 9, 2026



Rajarshi Mukherjee

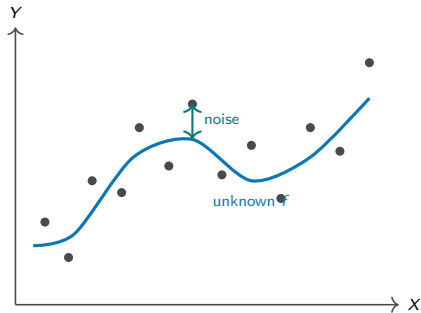


James M. Robins



Zixiao Wang

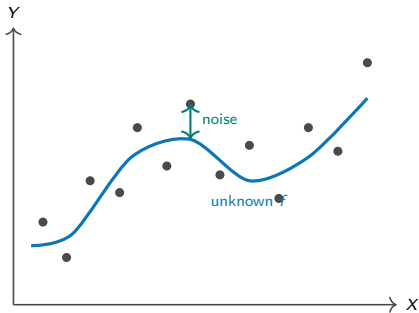
Setup



$$Y_i = f(X_i) + \varepsilon_i, \quad \text{Var}(\varepsilon_i | X_i) = \sigma^2.$$

The parameter of interest is the noise variance σ^2 ; f is a nuisance parameter.

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- $\sigma^2 = \mathbb{E}\{\text{Var}(Y | X)\}$ is a special case of an expected conditional covariance, connecting to causal inference problems.

Problem setting

We observe independent copies (X_i, Y_i) , $i = 1, \dots, n$, with $X_i \in [0, 1]^d$ and

$$\mathbb{E}(\varepsilon_i \mid X_i) = 0, \quad \mathbb{E}(\varepsilon_i^2 \mid X_i) = \sigma^2, \quad \mathbb{E}(\varepsilon_i^4 \mid X_i) \leq M.$$

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f is s -Hölder-smooth with $s > 1$.

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f is s -Hölder-smooth with $s > 1$.

Feature density

The density p satisfies $0 < \underline{p} \leq p \leq \bar{p}$; its Hölder smoothness is denoted β_p .

Problem setting

We observe independent copies (X_i, Y_i) , $i = 1, \dots, n$, with $X_i \in [0, 1]^d$ and

$$\mathbb{E}(\varepsilon_i \mid X_i) = 0, \quad \mathbb{E}(\varepsilon_i^2 \mid X_i) = \sigma^2, \quad \mathbb{E}(\varepsilon_i^4 \mid X_i) \leq M.$$

Regression smoothness

f is s -Hölder-smooth with $s > 1$.

Feature density

The density p satisfies $0 < \underline{p} \leq p \leq \bar{p}$; its Hölder smoothness is denoted β_p .

$$\boxed{s > 1, \quad d > 4s}, \text{ with } \sigma^2 \in [\underline{v}, \bar{v}].$$

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- ▶ **2026:** After our work, Aronow and Lopatto (arxiv v3) show that this rate is minimax optimal for the zero smoothness case $\beta_g = 0$.

Close pairs (Robins et al., 2008)

Since $Y_i = f(X_i) + \varepsilon_i$ and $Y_j = f(X_j) + \varepsilon_j$, we have $Y_i - Y_j = f(X_i) - f(X_j) + \varepsilon_i - \varepsilon_j$ and

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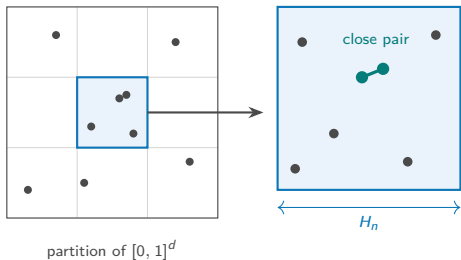
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Exploiting smoothness beyond one requires removing nonconstant polynomial trends.

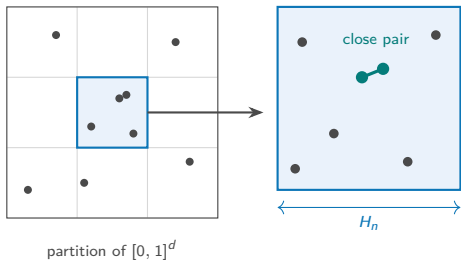
Two spatial scales



Coarse scale: project out the local trend

Cells have diameter of order $H_n \asymp n^{-1/d}$; use the points in a each cell to project out a polynomial.

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Fine scale: control the remainder

Inside a cell, choose a pair at distance $O(H_n \varepsilon_n)$, where $\varepsilon_n \rightarrow 0$.

The building block: squared contrasts

In a cell Q , collect the responses in Y_Q and the regression values in f_Q . As is standard, we start from contrasts using vectors a_Q based on only the covariates, with $\|a_Q\|_2 = 1$.

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Conditional bias

$$\mathbb{E}\{(a_Q^\top Y_Q)^2 \mid X_1, \dots, X_n\} = \underbrace{\sigma^2}_{\text{variance}} + \underbrace{(a_Q^\top f_Q)^2}_{\text{conditional bias}} .$$

Projecting out the polynomial trend

Rescale Q to $[0, 1]^d$, with coordinates U_i . Let $\psi(u)$ list all monomials of total degree at most ℓ , and let Φ_Q have rows $\psi(U_i)^\top$, where

$$\ell = \lceil s \rceil - 1, \quad q = \binom{d + \ell}{\ell}.$$

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For an eligible pair $e = (i, j)$, let $d_e = e_i - e_j$ in cell coordinates and define

$$R_Q = I - \Phi_Q(\Phi_Q^\top \Phi_Q)^{-1} \Phi_Q^\top, \quad a_Q = \frac{R_Q d_e}{\sqrt{d_e^\top R_Q d_e}}.$$

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Controlled correction

Since the polynomial basis is Lipschitz, the initial contrast is already small: $\Phi_Q^\top d_e = \psi(U_i) - \psi(U_j) = O(\varepsilon_n)$.

Screening cells

A simplified version of our estimator retains a cell Q only when:

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3. It contains a pair with

$$\|U_i - U_j\|_\infty \leq \varepsilon_n, \quad d_e^\top R_Q d_e \geq v_0 > 0.$$

The simplified estimator

Let $\mathcal{T}_n$ be the set of cells passing the screen, and write $K_n = |\mathcal{T}_n|$.

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Simplified two-scale estimator

$$\hat{\sigma}_{\text{simple}}^2 = \begin{cases} \frac{1}{K_n} \sum_{Q \in \mathcal{T}_n} (a_Q^\top Y_Q)^2, & K_n \geq c_0 n \varepsilon_n^d, \\ (\underline{v} + \bar{v})/2, & \text{otherwise.} \end{cases}$$

Both scales are needed

Write the regression values in a cell as a local polynomial plus remainder (bounded via local Taylor series expansion due to Hölder assumption):

$$f_Q = \Phi_Q \theta_Q + r_Q, \quad \|r_Q\|_\infty \lesssim H_n^s.$$

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Projected close pair bias bound

On a cell passing the screen,

$$|a_Q^\top f_Q| \lesssim H_n^s \varepsilon_n \quad \implies \quad \underbrace{\mathbb{E}\{(a_Q^\top Y_Q)^2 \mid X\} - \sigma^2}_{\text{bias}} \lesssim H_n^{2s} \varepsilon_n^2.$$

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Given all covariates, data in different cells is independent. Due to bounded fourth moments,

$$\text{Var}\left(\frac{1}{K_n} \sum_{Q \in \mathcal{T}_n} (a_Q^\top Y_Q)^2 \middle| X\right) \lesssim \frac{1}{K_n} \lesssim \frac{1}{n\varepsilon_n^d}.$$

The risk calculation

With $H_n \asymp n^{-1/d}$, the main contributions are

$$\text{MSE}(\hat{\sigma}_{\text{simple}}^2) \lesssim \underbrace{H_n^{4s} \varepsilon_n^4}_{\text{squared bias}} + \underbrace{\frac{1}{n \varepsilon_n^d}}_{\text{variance}},$$

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Examples of the rates

For illustration, take $d = 8$ and $s = 1.5$.

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The proof does not use density smoothness.

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Robins et al. (2008)

For $\beta_p > \beta_p^*$, the HOIF construction gives

$$\sup_{\theta} \mathbb{E}_{\theta} (\hat{\sigma}_{\text{HOIF}}^2 - \sigma^2)^2 \leq Cn^{-8s/(d+4s)},$$

under some regularity conditions.

Reusing the pairs: the all-pairs ridge extension

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One global stabilized extrapolation

Pool the M retained pairs and fit

$$(\hat{a}, \hat{b}) = \arg \min_{a, b} \left\{ \sum_e \{Z_e - a - b(s_e - \bar{s})\}^2 + \lambda M b^2 \right\}.$$

Estimate σ^2 by $\text{clip}_{[\underline{v}, \bar{v}]}(\hat{a} - \bar{s}\hat{b})$; use $\underline{v}$ if $M = 0$.

Simulation study

- ▶ **Methods:** simplified, all-pairs two-scale, local polynomial/affine
 - Tong–Wang (2005, Biometrika) extrapolation,
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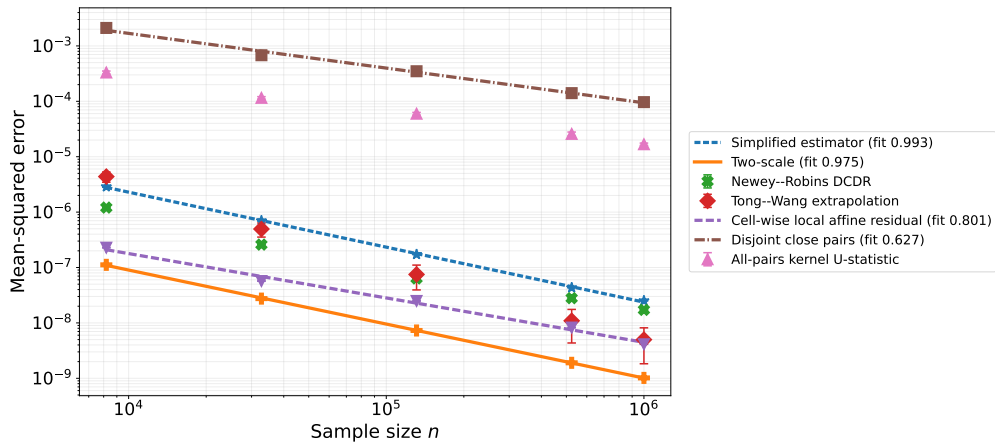
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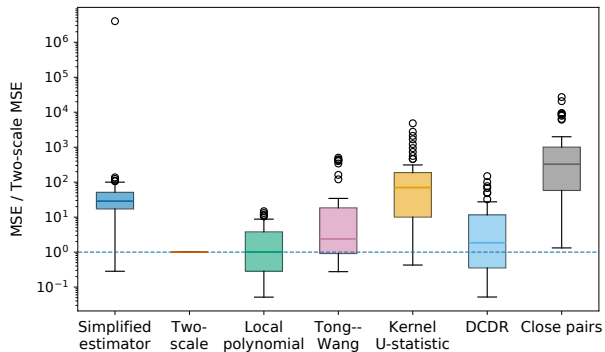
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- ▶ **Evaluation:** compute MSE for each estimator, then average over sampled designs.

Log-log MSE plot



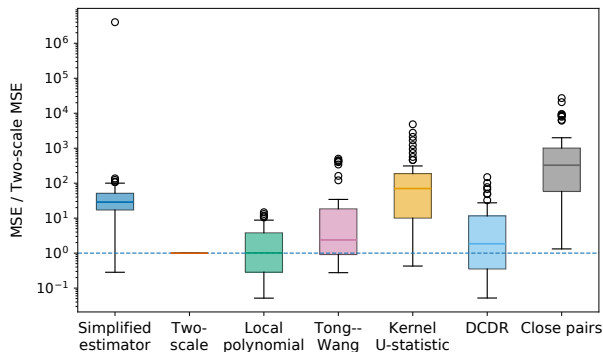
Uniform density, $d = 8$, $s = 1.5$.

Varying settings



Each ratio is a method's MSE divided by the all-pairs two-scale MSE.

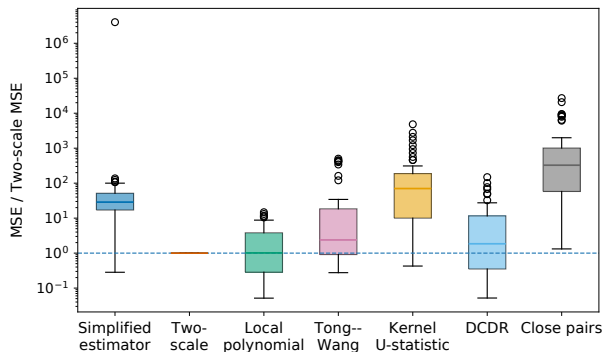
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Method	Geo. mean	Wins
Two-scale	1.00	15/48
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simplified	30.81	1/48
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Local residuals or DCDR can win for very smooth regression functions.

Empirical covariates with a controlled response model

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Method	Geometric mean relative MSE	Wins
All-pairs two-scale	1.00	14/14
Kernel U -statistic	4.02	0/14
Local polynomial	12.57	0/14
Tong–Wang	16.32	0/14
simplified	21.41	0/14
Close pairs	29.29	0/14
DCDR	51.55	0/14

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Human effort:

- ▶ simplify and clean up methods,
- ▶ find intuition,
- ▶ perform proper idea attribution (read related papers, make sure we understand the novelty),
- ▶ check arguments,
- ▶ write human-readable paper (not AI-style slop papers).

Some takeaways

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- ▶ Initial outputs can fail to pass reality/practicality checks (e.g., too many hyperparameters)
- ▶ Novelty should ultimately always be checked by humans, developing deep expertise is required in order to perform proper attribution.
- ▶ Thank you!

